

Daniel Foister, Jason Gun, Timyathus Spikes, Tyrese Dickson - DTJT Group
Shehenaz Shaik
CSCI 4350
Sprint #7 - 3/26-4/2

Roles:

- Product Owner: Jason Gun
- Scrum Master: Tyrese Dickson
- Development Team: Daniel Foister, Timyathus Spikes

Time Limit: Two Days (3/26, 3/31, 4/2)

Attendance: All group members were present this week.

Project Github Repo: <https://github.com/12061062/StockBuddy>

Sprint Goal:

- The goal for this sprint was to improve feedback loops and system reliability by enhancing test result visibility, implementing notifications for failed builds/tests, improving existing test cases based on past failures, and identifying and fixing recurring bugs. These improvements help ensure faster feedback, better debugging, and a more stable system over time.

PBIs/User Stories/To-Do Chosen for Sprint:

- **SCRUM-47:** Enhance test result output to improve readability and clarity for developers.
 - Estimation: 4 story points
- **SCRUM-48:** Implement a notification mechanism to alert developers when a build or test fails.
 - Estimation: 4 story points
- **SCRUM-49:** Improve existing test cases based on analysis of previous test failures.
 - Estimation: 4 story points
- **SCRUM-50:** Identify and resolve recurring defects within the system.
 - Estimation: 4 story points
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- The PBIs/tasks selected for this sprint were based on our team's average velocity of around 16 story points per sprint from previous sprints. These features were prioritized because they improve feedback loops, help identify issues faster, and enable the team to improve the system based on past failures continuously.

Sprint Planning

- During sprint planning, our team reviewed the backlog and selected/created four PBIs/tasks that aligned well with StockBuddy's current development stage. The tasks were distributed evenly among our team members, and the acceptance criteria were clarified before beginning implementation.

Total Sprint Duration:

- 4 members x 7 hours/week x 1 Week/Sprint = **32 man-hours available**

Tasks Performed:

- Task Name: Enhance test result output to improve readability and clarity for developers.
- Task Owner: Tyrese Dickson

- Planned Duration: 5 hours
- Actual Duration: 6 hours
- Status: Completed
- Challenges: None
- Lessons Learned: None

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- Task Name: Implement a notification mechanism to alert developers when a build or test fails.
- Task Owner: Timyathus Spikes
- Planned Duration: 5 hours
- Actual Duration: 6 hours
- Status: Completed
- Challenges: Making sure the notifications are triggered correctly for all failure cases.
- Lessons Learned: Immediate notifications improve response time to issues.

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- Task Name: Improve existing test cases based on analysis of previous test failures.
- Task Owner: Jason Gun
- Planned Duration: 5 hours
- Actual Duration: 6 hours
- Status: Completed
- Challenges: None
- Lessons Learned: Improving test coverage reduced recurring bugs.

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- Task Name: Identify and resolve recurring defects within the system.
- Task Owner: Daniel Foister
- Planned Duration: 7 hours
- Actual Duration: 8 hours
- Status: Completed
- Challenges: Tracking down the root causes of recurring bugs required additional debugging.
- Lessons Learned: None

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Sprint Review:

- Was the Sprint Goal Achieved?:
 - The Sprint Goal was Achieved.
 - Our DoD: Our work/code maps to all requirements specified in the Jira Board (backlog).
 - Our team successfully improved test result visibility, implemented notifications for failed builds/tests, improved test cases based on previous failures, and resolved recurring system defects. These improvements strengthen feedback loops and the overall reliability/stability of StockBuddy.
- Focus for the Next Sprint:
 - Security

- PO/Stakeholder Review:
 - The Product Owner reviewed all completed tasks/backlog items:
 - The test results are clearly displayed and easy to understand.
 - Notifications are triggered when builds or tests fail.
 - Test cases have been improved to better capture edge cases.
 - Recurring bugs have been identified and resolved.
 - Feedback Provided:
 - Continue expanding test coverage.
 - Maybe improve logging for easier debugging.
 - Priority Revisions:
 1. Additional test improvements.
 2. Performance improvements moved higher in priority.
- Testing
 - During this sprint, we improved existing tests and feedback mechanisms to ensure developers can quickly identify and fix issues. These improvements helped verify system reliability and reduce recurring defects.
 - Unit Tests:
 - Test that the cart tax is calculated correctly based on the items in the cart.
 - Test that an email is sent successfully after checkout.
 - Test that role-based authentication correctly restricts access to protected features.
 - Test that scanned items are added correctly even when barcode input is inconsistent or incomplete.

Sprint Retrospective:

- Changes in the Scrum Process:
 - Again, our team does not believe we need to make any changes to the Scrum Process yet, in the context of this Sprint.
- Start: Improving test coverage based on past failures.
- Stop: Ignoring minor recurring bugs.
- Continue: Frequent push requests/commits, clear communication, and consistent task tracking.
- Other Planned Adjustments/Concerns:
 - No major concerns were identified during this sprint.
- Team Velocity:
 - Sprint 1: 12 story points
 - Sprint 2: 16 story points
 - Sprint 3: 18 story points
 - Sprint 4: 17 story points
 - Sprint 5: 17 story points
 - Sprint 6: 16 story points
 - Sprint 7: 16 story points
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 - Average Velocity: ~16 story points per sprint

Backlog Grooming

- PBIs/Tasks Added/Removed/Reprioritized
 - As a system, I want to securely store user credentials and protect sensitive data so that unauthorized access and data exposure are prevented.
 - As a developer, I want to validate user input and enforce role-based access control so that the system is protected from malicious input and unauthorized actions.
 - Hash and securely store user credentials instead of using plain text or hardcoded values.
 - Sanitize input fields to reduce the risk of injection attacks.
 - Ensure sensitive data is not exposed in logs or error messages.
 - Reinforce role-based access control across all system features.
- Definition of Ready (DoR)
 - Our DoR: All backlog items/tasks selected for the next sprint meet our defined acceptance criteria, estimated story points, identified dependencies, and can be reasonably completed within one sprint.

Daily Standup #1:

- Duration: 5 Minutes
- Facilitator: Tyrese Dickson

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Tyrese Dickson

1. **What did I do yesterday?**

I wrapped up working on implementing the updates to the GitHub repository with an automated testing file.

2. **What do I plan to do today?**

Today, I planned out how I am going to enhance the test result output to improve readability and clarity for developers.

3. **What are any challenges or impediments that I have experienced?**

None

Timyathus Spikes

1. **What did I do yesterday?**

I wrapped up working on implementing the automated testing on GitHub Actions.

2. **What do I plan to do today?**

Today, I planned out how I am going to implement a notification mechanism to alert developers when a build or test fails.

3. **What are any challenges or impediments that I have experienced?**

None

Jason Gun

1. **What did I do yesterday?**

I wrapped up working on improving the workflow so code changes are tested immediately after development.

2. **What do I plan to do today?**

Today, I planned out how I am going to improve existing test cases based on the analysis of previous test failures.

3. **What are any challenges or impediments that I have experienced?**

None

Daniel Foister

1. **What did I do yesterday?**

I wrapped up working on reducing the number of steps required to run and test the app.

2. **What do I plan to do today?**

Today, I planned out how I am going to identify and resolve recurring defects within the system.

3. **What are any challenges or impediments that I have experienced?**

None

Daily Standup #2:

- Duration: 5 Minutes
- Facilitator: Tyrese Dickson

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Tyrese Dickson

4. **What did I do yesterday?**

Yesterday, I planned out how I was going to enhance the test result output to improve readability and clarity for developers.

5. **What do I plan to do today?**

Today, I am going to begin enhancing the test result output to improve readability and clarity for developers, which I planned out during the last session.

6. **What are any challenges or impediments that I have experienced?**

None

Timyathus Spikes

4. **What did I do yesterday?**

Yesterday, I planned out how I was going to implement a notification mechanism to alert developers when a build or test fails.

5. **What do I plan to do today?**

Today, I am going to begin implementing a notification mechanism to alert developers when a build or test fails, which I planned out during the last session.

6. **What are any challenges or impediments that I have experienced?**

None

Jason Gun

4. **What did I do yesterday?**

Yesterday, I planned out how I was going to improve existing test cases based on the analysis of previous test failures.

5. **What do I plan to do today?**

Today, I am going to begin improving existing test cases based on the analysis of previous test failures, which I planned out during the last session.

6. **What are any challenges or impediments that I have experienced?**

None

Daniel Foister

4. **What did I do yesterday?**

Yesterday, I planned out how I was going to identify and resolve recurring defects within the system.

5. **What do I plan to do today?**

Today, I am going to begin identifying and resolving recurring defects within the system, which I planned out during the last session.

6. **What are any challenges or impediments that I have experienced?**

None

Daily Standup #3:

- Duration: 5 Minutes
- Facilitator: Tyrese Dickson

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Tyrese Dickson

7. **What did I do yesterday?**

I continued working on enhancing the test result output to improve readability and clarity for developers.

8. **What do I plan to do today?**

Complete my assigned task and plan for Sprint #8.

9. **What are any challenges or impediments that I have experienced?**

None

Timyathus Spikes

7. **What did I do yesterday?**

I continued working on implementing a notification mechanism to alert developers when a build or test fails.

8. **What do I plan to do today?**

Complete my assigned task and plan for Sprint #8.

9. **What are any challenges or impediments that I have experienced?**

None

Jason Gun

7. **What did I do yesterday?**

I continued working on improving existing test cases based on the analysis of previous test failures.

8. **What do I plan to do today?**

Complete my assigned task and plan for Sprint #8.

9. **What are any challenges or impediments that I have experienced?**

None

Daniel Foister

7. **What did I do yesterday?**

I continued working on identifying and resolving recurring defects within the system.

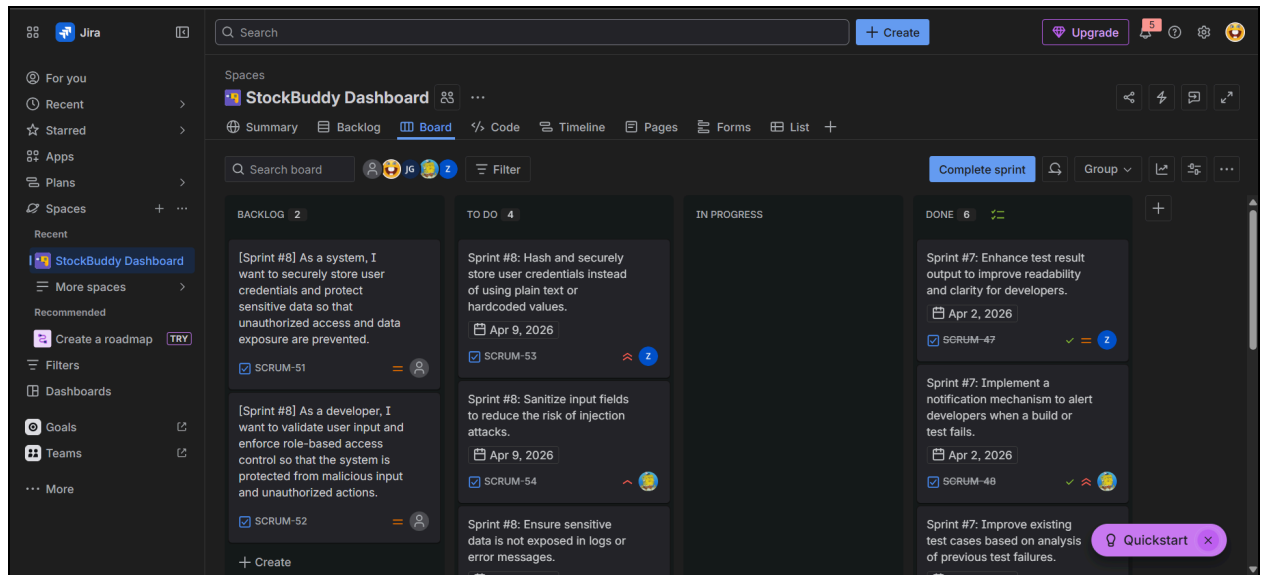
8. **What do I plan to do today?**

Complete my assigned task and plan for Sprint #8.

9. **What are any challenges or impediments that I have experienced?**

None

Exported Copy of Scrum Board:



END.